

Roll No.

Total Pages : 5

11051/NJ**D-17/2111****APPLIED PHYSICS**

Paper-1101T

Semester-I

Time Allowed : 3 Hours] [Maximum Marks : 50

Note : The candidates are required to attempt **three** questions each from Sections A and B carrying 5 marks each and the entire Section C consisting of 10 short answer type questions carrying 2 marks each.

SECTION—A

1. Define the Amplitude and period of S.H.M. Show that the total energy for a particle executing SHM remains constant. 2,3

2. What do you mean by resolution and resolving power of an Optical instrument? Deduce an expression for the resolving power of a Telescope. 2,3

3. Under what condition, damped motion becomes oscillatory? Discuss in detail the damped oscillations in an LCR circuit. 1,4

4. Calculate the minimum number of lines in a grating which will just resolve the sodium lines of wavelengths $5890 \text{ \AA}$ and $5896 \text{ \AA}$ in the first order spectrum. 5

5. Explain the set up and working of Michelson interferometer. Discuss, how it can be used to determine the thickness of transparent plate. 2,3

SECTION—B

6. What do you understand by the terms stimulated absorption, spontaneous emission and stimulated

emission? Derive the relation between transition probabilities of spontaneous and stimulated emission. 2,3

7. Discuss the various components of Laser. Explain the neat diagram, principle and working of He-Ne laser. 2,3

8. What is an optical fiber? Define and explain the terms : Acceptance angle and Numerical aperture. Calculate the critical angle and numerical aperture of a step index fiber if the refractive indices for its core and cladding are 1.52 and 1.41 respectively. 1,2,3

9. What do you mean by time dependent Schrödinger equation for a free particle? Discuss the interpretation of position probability density and normalization wave function. 3,2

10. Derive an expression for the wave function and energy of a particle in one-dimensional infinite potential box using the Schrödinger wave equation.

5

SECTION—C

11. Write short notes on the following : 10×2=20

- (i) What is an expectation value of momentum operator?
- (ii) Explain the Physical significance of ψ and $|\psi|^2$.
- (iii) What is lightly damped oscillator?
- (iv) Write two differences between the Fraunhofer and Fresnel diffraction.
- (v) On what factors does the resolving power of grating depend?

- (vi) Distinguish between linearly, circularly and elliptically polarized light.
- (vii) Discuss the salient characteristics of Laser.
- (viii) Write time independent Schrödinger wave equation.
- (ix) Explain the term quality factor with reference to a under damped oscillator.
- (x) What is cause of signal attenuation in Fiber communication?

SECTION—A

1. Derive the differential equation of a damped electrical oscillator and find the general solution for this differential equation. Also discuss the case of lightly damped oscillations. 3,2

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2. Discuss the Lissajous figures formed by the superposition of two simple harmonic motions for different values of phase difference. 5
3. Describe the principle, construction and working of Michelson's interferometer. 5

4. What do you mean by resolving power? Explain the Rayleigh's criterion for resolution. Derive the expression for the resolving power of a telescope. 1,1,3
5. Distinguish between linearly, circularly and elliptically polarized light giving mathematical expressions for them. 5

SECTION—B

6. Explain the construction and working of Helium-Neon laser with labeled diagrams. In He-Ne laser why is it necessary to use narrow tubes? 4,1
7. What are Einstein's coefficients? Derive Einstein's relations. 2,3

8. What is meant by Acceptance angle for an optical fibre? Derive the expression for the numerical Aperture and Acceptance angle of an optical fibre.

2,3

9. Derive the Schrödinger's time dependent and time independent wave equations.

5

10. Solve the Schrödinger's equation for one dimensional motion of a particle in a box of length L and show that its Eigen value is inversely proportional to the square of length L .

5

SECTION—C

11. Write short notes on the following : $10 \times 2 = 20$

~~(a)~~ What is Quality factor of an oscillator?

~~(b)~~ What is simple harmonic motion?

~~(c)~~ Are all periodic motions necessarily vibratory motions? Explain briefly.

(d) What do you understand by Coherent sources of light?

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(e) What is the difference between O-ray and E-ray?

(f) What is population inversion?

(g) What is Optical resonant cavity?

(h) What is the difference between step index and graded-index fibre?

(i) What is the physical significance of wave function?

- (h) What is the difference between step index and graded-index fibre?
- (i) What is the physical significance of wave function?
- (j) Calculate the de-Broglie wavelength of an electron accelerated by a potential difference of 100 volts.